Quantitative evaluation of activation maps for weakly-supervised lung nodule segmentation

Authors: Finn Behrendt, Suyash Sonawane, Debayan Bhattacharya, Lennart Maack, Julia Krüger, Roland Opfer, and Alexander Schlaefer

Students: RAMI Ilias & SAROUI Rachida [Group:22]

Motivation

•Manual Assessment Challenge: Assessing chest radiographs manually is time-consuming, error-prone, and demands expert knowledge [1].
•Deep Learning Potential: Deep learning models enhance diagnostic accuracy and reduce radiologists' workload [2]. Despite their success, deep learning models often lack transparent decision-making processes, limiting clinical trust.
•Explainability Gap: To improve model explainability, techniques such as Grad-CAM [6] have been introduced. These methods generate heatmaps to highlight the regions influencing predictions. While helpful, Grad-CAM often struggles with precise localization.
•ViTs as a Solution: Vision Transformers (ViTs) leverage self-attention mechanisms, offering improved visualization of relevant regions and enhancing model interpretability [4].

Key Question

Does high-performing classification models focus on the correct regions to detect pulmonary nodules?

Objectif

This work [3] aims to evaluate the accuracy of activation maps for pulmonary nodule segmentation, addressing the gap between classification success and meaningful visual explanations.

CXR's Image Classification

DataSet : Node21

This dataset consists of frontal chest radiographs with annotated bounding boxes around nodules. It consists of 4882 frontal chest radiographs where 1134 CXR images (1476 nodules) are annotated with bounding boxes around nodules and the remaining 3748 images are free of nodules hence represent the negative class.

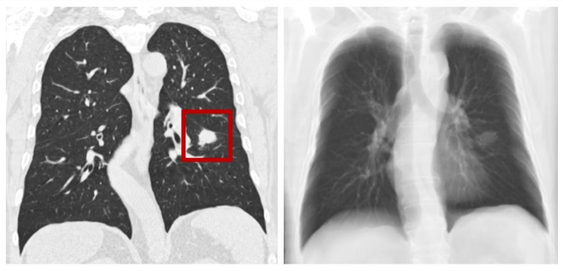

Fig 1: Example from the Node21 dataset

Architectures Overview :

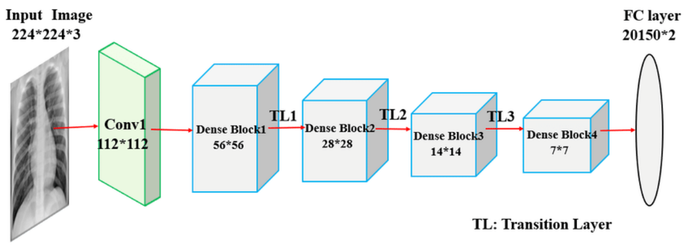

Fig 2: DenseNet121 Architecture Overview

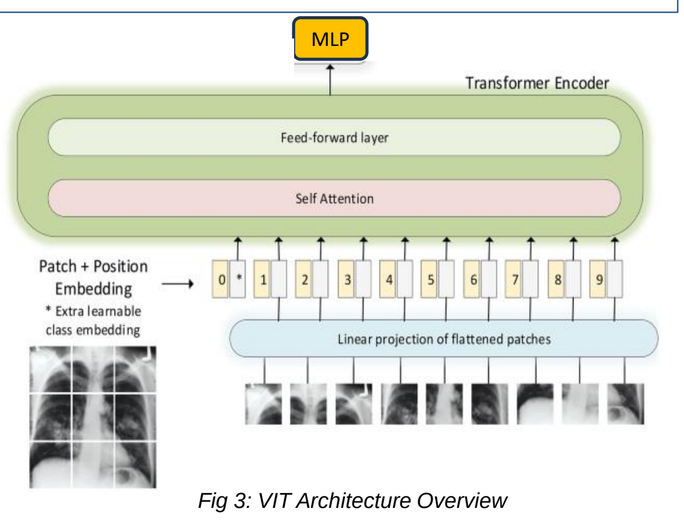

Fig 3: VIT Architecture Overview

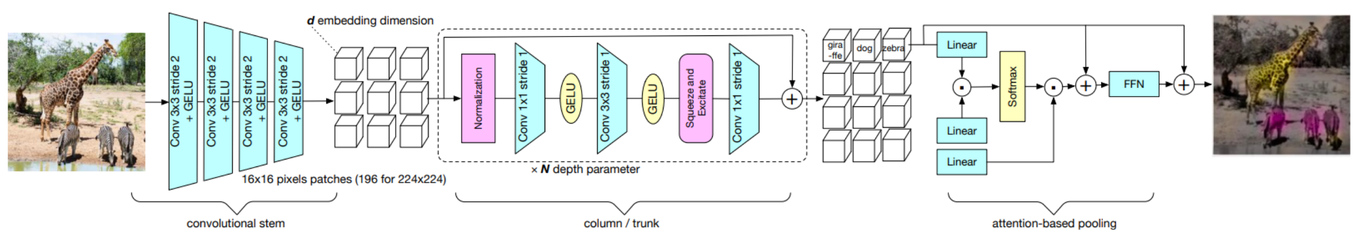

Fig 4: PatchConvNet Architecture Overview

Method Overview

Method Overview using Grad-CAM :

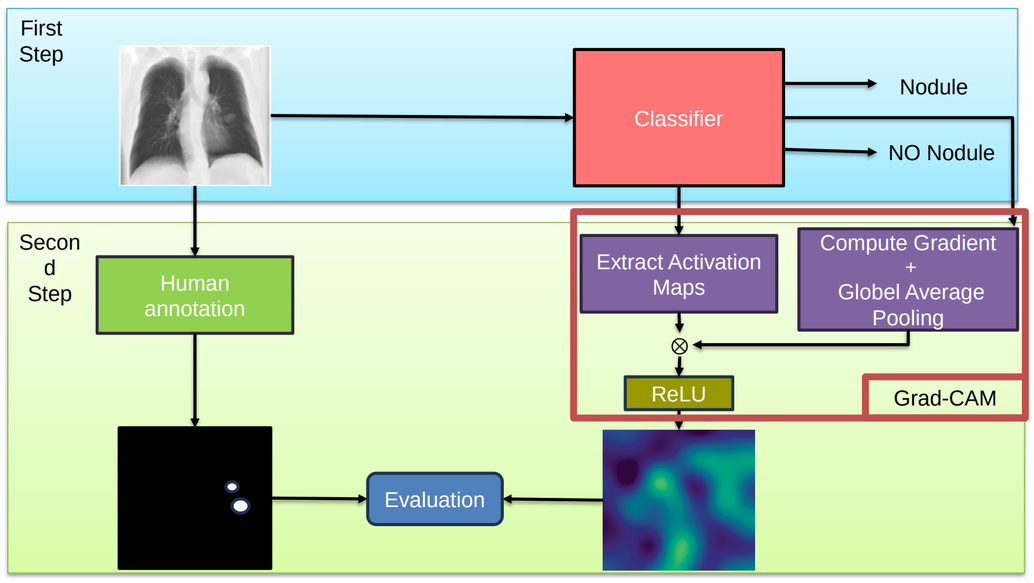

Fig 5: Method Overview using Grad-CAM

Method Overview using Attention Maps for Transformer-based Models :

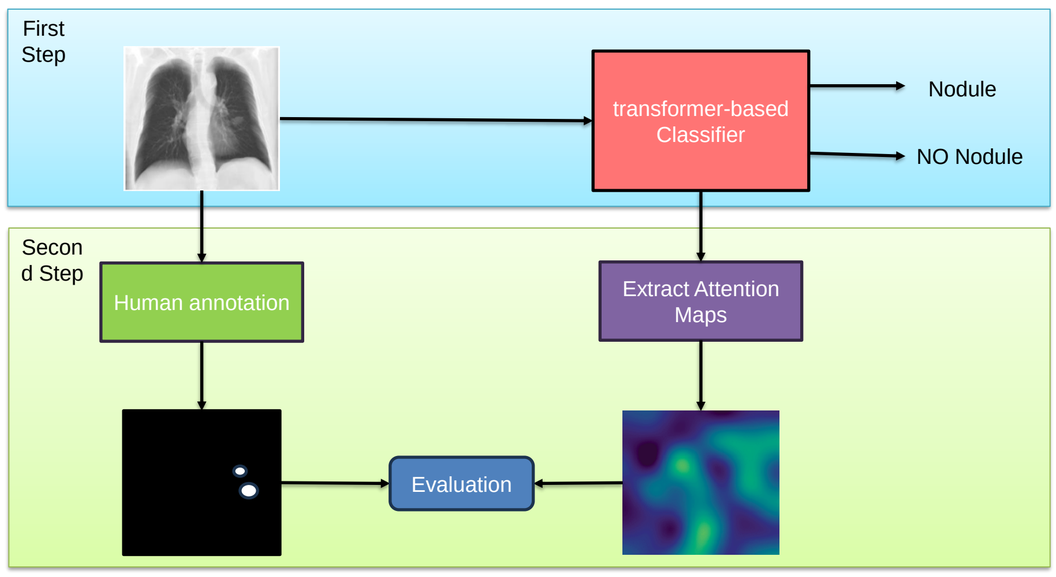

Fig 6: Method Overview using Attention Maps for Transformer-based Models

Results

Visualization of Activation Maps :

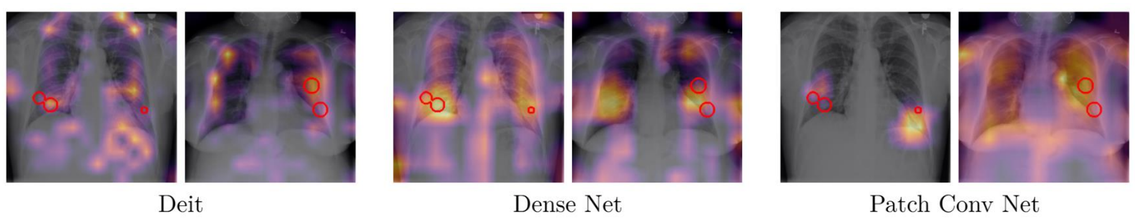

Fig 7: example of visualization of Activation Map Across Different Network Architectures ---red circles are the GT and the heat map is the attention scores or heat map generated using Grad-CAM---

Performances :

Model	AUPRC	AUROC
Deit	80.93 ± 0.90	81.42 ± 1.08
Dense Net	75.50 ± 0.64	76.89 ± 0.89
Patch Conv Net	74.67 ± 3.14	77.56 ± 2.64

Table 1:Nodule classification Performance

	AUPRC	
Model	GradCAM	Attention
Deit	3.45 ± 2.38	3.42 ± 1.36
Dense Net	1.36 ± 0.34	N/A
Patch Conv Net	2.51 ± 1.57	5.9 ± 0.92

Table 2:Nodule segmentation Performance

Conclusions

Despite high classification accuracy, the models' activation and attention maps poorly corresponded to actual nodule locations, indicating reliance on contextual cues rather than direct identification. This underscores that high classification performance does not necessarily equate to accurate localization in weakly-supervised settings.

References

1. Brady, A. P., "Error and discrepancy in radiology: inevitable or avoidable?," Insights into Imaging 8, 171–182 (Feb. 2017)
2. Çallı, E., Sogancioglu, E., van Ginneken, B., van Leeuwen, K. G., and Murphy, K., "Deep learning for chest x-ray analysis: A survey," Med. Image Anal. 72, 102125 (2021).
3. Behrendt, F., Sonawane, S., Bhattacharya, D., Maack, L., Krüger, J., Opfer, R., & Schlaefer, Quantitative evaluation of activation maps for weakly-supervised lung nodule segmentation.
4. Behrendt, F., Bhattacharya, D., Krüger, J., Opfer, R., and Schlaefer, A., "Data-efficient vision transformers for multi-label disease classification on chest radiographs," Current Directions in Biomedical Engineering 8(1), 34–37 (2022).
5. E. Sogancioglu et al., "Nodule Detection and Generation on Chest X-Rays: NODE21 Challenge," IEEE Transactions on Medical Imaging, vol. 43, no. 8, pp. 2839–2854, Aug. 2024
6. Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D., "Grad-cam: Visualexplanations from deep networks via gradient-based localization," in [Proceedings of the IEEE internationalconference on computer vision], 618–626 (2017)

© Poster Template by Genigraphics® 1.800.790.4001 www.genigraphics.com